

Sarvasetty Revanth Kumar

+91 7416476715 | sarvasettyrevanth.66@gmail.com | LinkedIn | Github

Profile Summary

Computer Science and Engineering (Data Science) graduate with hands-on experience building machine learning applications involving anomaly detection, NLP, multimodal classification, and model inference. Experienced with Python, scikit-learn, XGBoost, Transformers/RoBERTa, Pandas, NumPy, FastAPI, PostgreSQL, and MLflow. Built ML pipelines covering feature preparation, model evaluation, anomaly scoring, classification, and API-based inference.

Technical Skills

Programming: Python, SQL

Machine Learning: Scikit-learn, XGBoost, Anomaly Detection, Classification, Feature Engineering, Model Evaluation, Hyperparameter Tuning

Deep Learning & NLP: PyTorch, Transformers, RoBERTa, NLP, Text Classification, Multimodal Learning

Time-Series & Statistical Modelling: EWMA, GJR-GARCH, HMM, Volatility Forecasting, VaR, Expected Shortfall, Statistical Backtesting

Data: Pandas, NumPy, SciPy, Data Preprocessing, Exploratory Data Analysis, Statistical Analysis

ML Engineering: ML Pipelines, Model Inference, FastAPI, REST APIs, MLflow, Docker

Databases & Systems: PostgreSQL, Redpanda, GitHub

Projects

Adaptive Volatility Forecasting, Tail-Risk & Portfolio Decision Engine (AVF-TRPDE)

Python | Pandas | NumPy | XGBoost | HMM | GJR-GARCH | EWMA

- Built a walk-forward volatility forecasting engine across 4 models and 395 OOS folds, producing 1,580 predictions with fold-local training and leakage controls.
- Combined HMM regime probabilities with XGBoost and evaluated forecasting differences using DM tests, 500-bootstrap CIs, and 90% Model Confidence Sets.
- Implemented VaR 95/99%, Expected Shortfall, Kupiec/Christoffersen tests, volatility targeting, transaction costs, and stress testing, with 294-observation risk backtesting.
- Validated regime robustness across 3 assets × 119 OOS observations, showing Regime-XGBoost outperformed baseline XGBoost on 39.5–49.6% of observations rather than consistently improving forecasts.

Multimodal Cyberbullying Detection System

Python | Transformers | RoBERTa | FastAPI | Streamlit | FFmpeg

- Developed a multimodal classification pipeline for detecting abusive content across text, audio, and video inputs.
- Fine-tuned a RoBERTa Transformer model for abusive-content classification and integrated its predictions into the multimodal inference workflow
- Implemented video frame extraction, audio extraction, and speech-to-text processing to convert multimedia inputs into features for downstream classification.
- Evaluated the system using Accuracy, Precision, Recall, and F1-score; Dataset 1 reported 97.70% Accuracy, 95.00% Precision, 96.00% Recall, and 96.54% F1-score.

Education

Mohan Babu University, Tirupati

B.Tech in Computer Science and Engineering (Data Science) | 2026 | CGPA: 8.15/10